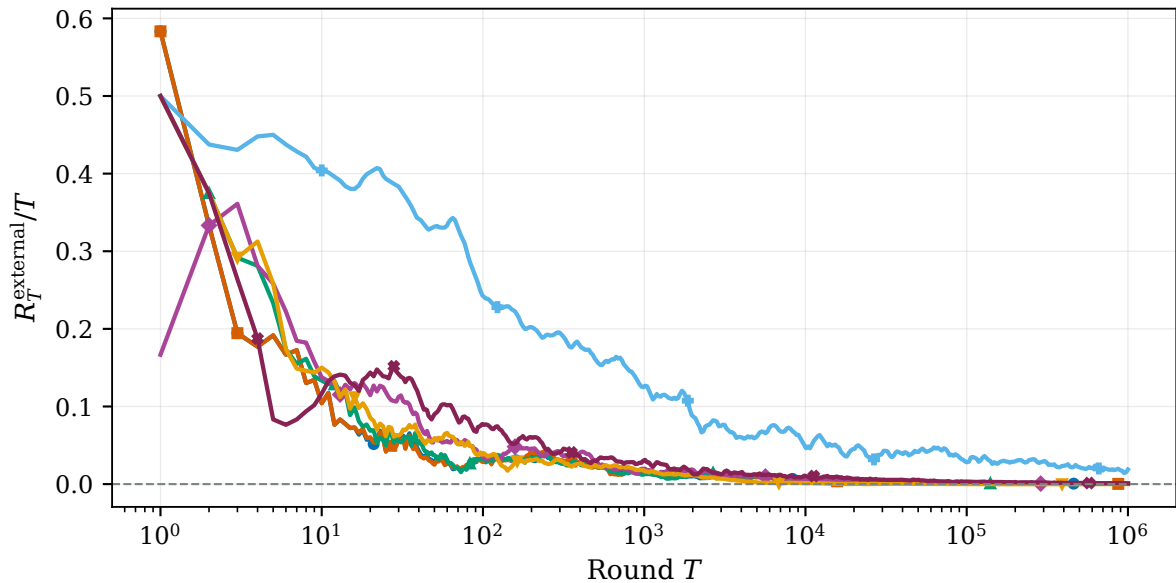


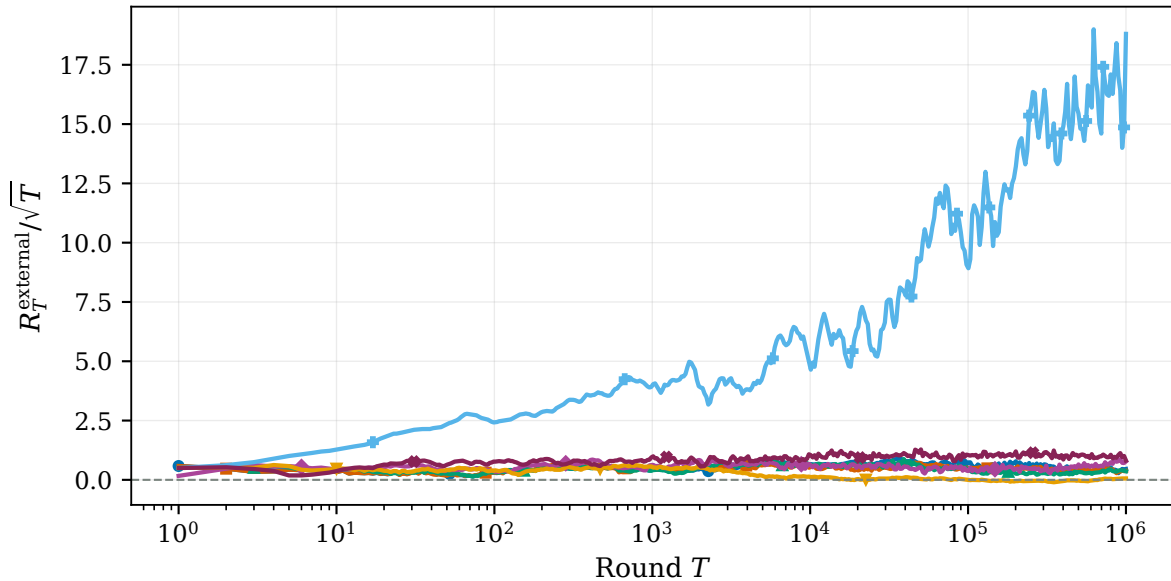
Experiment Information

Figure:	Algorithm comparison
Game:	Rock–Paper–Scissors–Lizard–Spock
Player:	0
Feedback:	Full information
Algorithms:	BM-Hedge vs BM-Hedge, BM-OptHedge vs BM-OptHedge, Hedge vs Hedge, Ito-Hedge vs Ito-Hedge, OptHedge vs OptHedge, RM vs RM, SRM vs SRM
Horizon:	1,000,000
Replicates:	12
Seed:	42
Metric:	External regret
View:	R / T
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
BM-Hedge vs BM-Hedge:	mean = 0.0004585, SD = 0.000417882, SE = 0.000120632
BM-OptHedge vs BM-OptHedge:	mean = 0.000377708, SD = 0.00028659, SE = 8.27313e-05
Hedge vs Hedge:	mean = 0.0003615, SD = 0.000232895, SE = 6.7231e-05
Ito-Hedge vs Ito-Hedge:	mean = 0.000776292, SD = 0.000338429, SE = 9.76961e-05
OptHedge vs OptHedge:	mean = 5.84167e-05, SD = 0.000230925, SE = 6.66622e-05
RM vs RM:	mean = 0.0188036, SD = 0.00634457, SE = 0.00183152
SRM vs SRM:	mean = 0.000813167, SD = 0.00026805, SE = 7.73793e-05



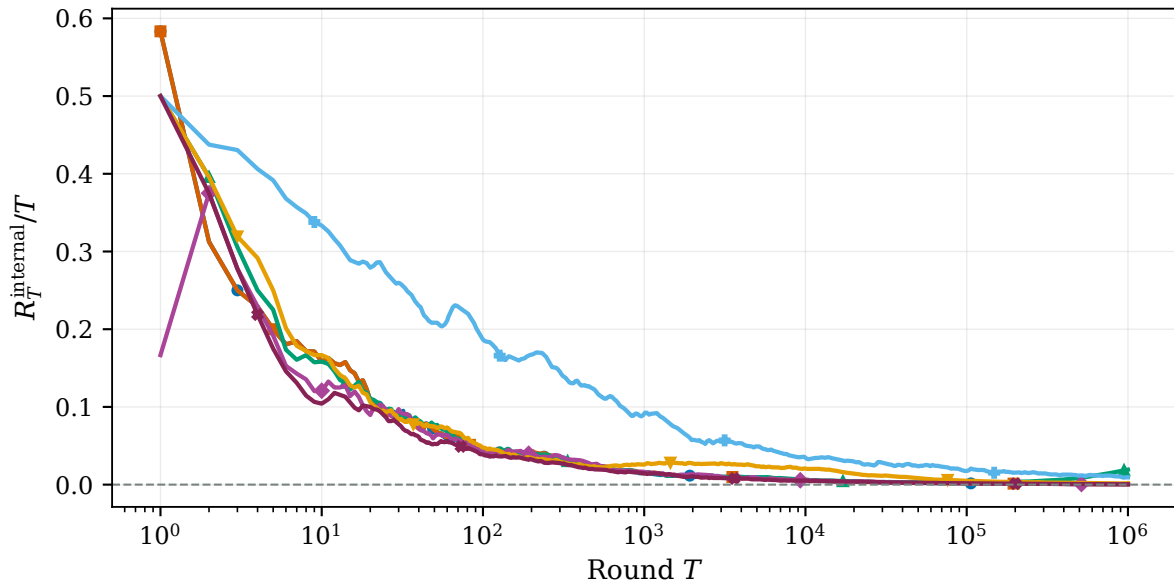
Experiment Information

Figure:	Algorithm comparison
Game:	Rock–Paper–Scissors–Lizard–Spock
Player:	0
Feedback:	Full information
Algorithms:	BM-Hedge vs BM-Hedge, BM-OptHedge vs BM-OptHedge, Hedge vs Hedge, Ito-Hedge vs Ito-Hedge, OptHedge vs OptHedge, RM vs RM, SRM vs SRM
Horizon:	1,000,000
Replicates:	12
Seed:	42
Metric:	External regret
View:	$R / \sqrt{T}$
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
BM-Hedge vs BM-Hedge:	mean = 0.4585, SD = 0.417882, SE = 0.120632
BM-OptHedge vs BM-OptHedge:	mean = 0.377708, SD = 0.28659, SE = 0.0827313
Hedge vs Hedge:	mean = 0.3615, SD = 0.232895, SE = 0.067231
Ito-Hedge vs Ito-Hedge:	mean = 0.776292, SD = 0.338429, SE = 0.0976961
OptHedge vs OptHedge:	mean = 0.0584167, SD = 0.230925, SE = 0.0666622
RM vs RM:	mean = 18.8036, SD = 6.34457, SE = 1.83152
SRM vs SRM:	mean = 0.813167, SD = 0.26805, SE = 0.0773793



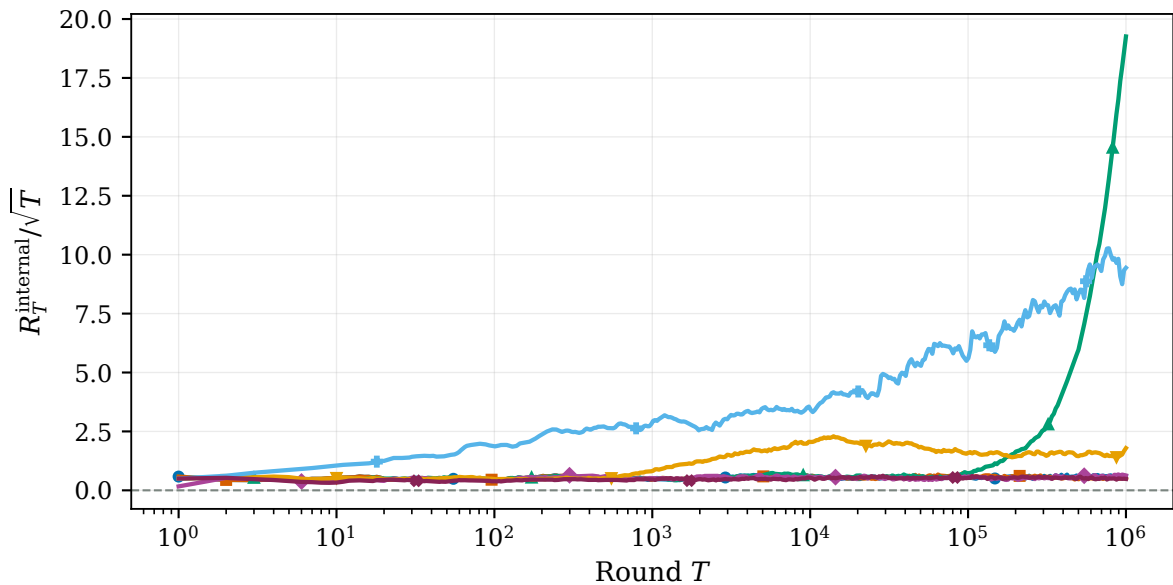
Experiment Information

Figure:	Algorithm comparison
Game:	Rock–Paper–Scissors–Lizard–Spock
Player:	0
Feedback:	Full information
Algorithms:	BM-Hedge vs BM-Hedge, BM-OptHedge vs BM-OptHedge, Hedge vs Hedge, Ito-Hedge vs Ito-Hedge, OptHedge vs OptHedge, RM vs RM, SRM vs SRM
Horizon:	1,000,000
Replicates:	12
Seed:	42
Metric:	Internal regret
View:	R / T
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
BM-Hedge vs BM-Hedge:	mean = 0.000606542, SD = 0.000300269, SE = 8.66802e-05
BM-OptHedge vs BM-OptHedge:	mean = 0.00052775, SD = 0.000151004, SE = 4.35911e-05
Hedge vs Hedge:	mean = 0.0192602, SD = 0.0115869, SE = 0.00334486
Ito-Hedge vs Ito-Hedge:	mean = 0.000629417, SD = 0.000139836, SE = 4.03671e-05
OptHedge vs OptHedge:	mean = 0.001791, SD = 0.000569282, SE = 0.000164338
RM vs RM:	mean = 0.00944167, SD = 0.0024986, SE = 0.000721285
SRM vs SRM:	mean = 0.000483333, SD = 0.000114538, SE = 3.30643e-05



Experiment Information

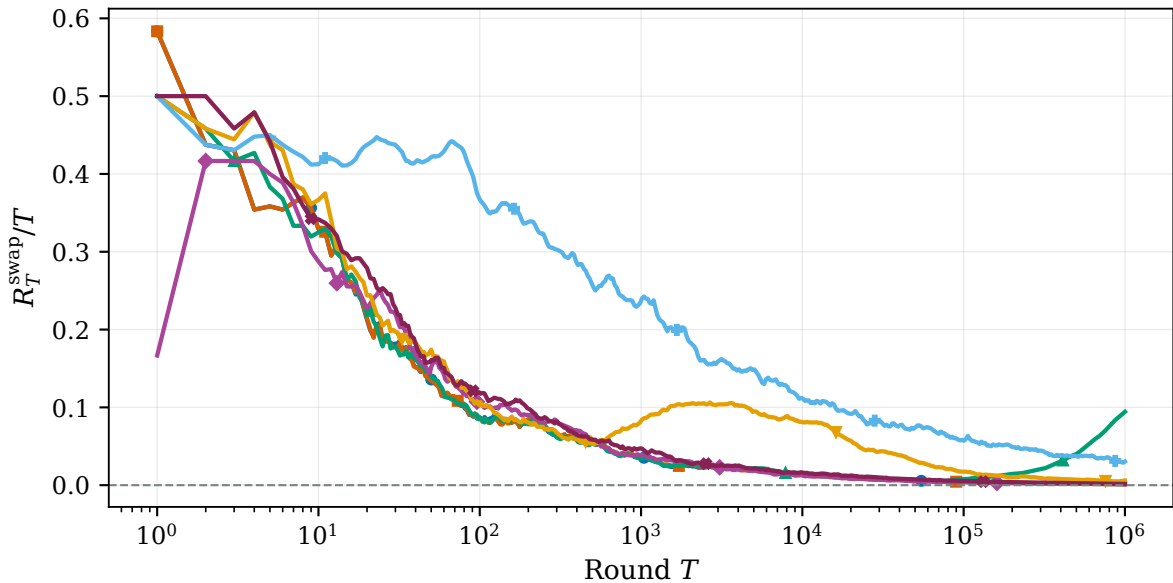
Figure:	Algorithm comparison
Game:	Rock–Paper–Scissors–Lizard–Spock
Player:	0
Feedback:	Full information
Algorithms:	BM-Hedge vs BM-Hedge, BM-OptHedge vs BM-OptHedge, Hedge vs Hedge, Ito-Hedge vs Ito-Hedge, OptHedge vs OptHedge, RM vs RM, SRM vs SRM
Horizon:	1,000,000
Replicates:	12
Seed:	42
Metric:	Internal regret
View:	$R / \sqrt{T}$
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
BM-Hedge vs BM-Hedge:	mean = 0.606542, SD = 0.300269, SE = 0.0866802
BM-OptHedge vs BM-OptHedge:	mean = 0.52775, SD = 0.151004, SE = 0.0435911
Hedge vs Hedge:	mean = 19.2602, SD = 11.5869, SE = 3.34486
Ito-Hedge vs Ito-Hedge:	mean = 0.629417, SD = 0.139836, SE = 0.0403671
OptHedge vs OptHedge:	mean = 1.791, SD = 0.569282, SE = 0.164338
RM vs RM:	mean = 9.44167, SD = 2.4986, SE = 0.721285
SRM vs SRM:	mean = 0.483333, SD = 0.114538, SE = 0.0330643



- BM-Hedge vs BM-Hedge
- BM-OptHedge vs BM-OptHedge
- Hedge vs Hedge
- Ito-Hedge vs Ito-Hedge
- OptHedge vs OptHedge
- RM vs RM
- SRM vs SRM

Experiment Information

Figure:	Algorithm comparison
Game:	Rock–Paper–Scissors–Lizard–Spock
Player:	0
Feedback:	Full information
Algorithms:	BM-Hedge vs BM-Hedge, BM-OptHedge vs BM-OptHedge, Hedge vs Hedge, Ito-Hedge vs Ito-Hedge, OptHedge vs OptHedge, RM vs RM, SRM vs SRM
Horizon:	1,000,000
Replicates:	12
Seed:	42
Metric:	Swap regret
View:	R / T
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
BM-Hedge vs BM-Hedge:	mean = 0.00125112, SD = 0.00058951, SE = 0.000170177
BM-OptHedge vs BM-OptHedge:	mean = 0.00119096, SD = 0.000271489, SE = 7.8372e-05
Hedge vs Hedge:	mean = 0.0942644, SD = 0.0577823, SE = 0.0166803
Ito-Hedge vs Ito-Hedge:	mean = 0.00148221, SD = 0.000305174, SE = 8.80963e-05
OptHedge vs OptHedge:	mean = 0.00593767, SD = 0.0027385, SE = 0.000790536
RM vs RM:	mean = 0.030369, SD = 0.00724807, SE = 0.00209234
SRM vs SRM:	mean = 0.00155679, SD = 0.000394619, SE = 0.000113917



Experiment Information

Figure:	Algorithm comparison
Game:	Rock–Paper–Scissors–Lizard–Spock
Player:	0
Feedback:	Full information
Algorithms:	BM-Hedge vs BM-Hedge, BM-OptHedge vs BM-OptHedge, Hedge vs Hedge, Ito-Hedge vs Ito-Hedge, OptHedge vs OptHedge, RM vs RM, SRM vs SRM
Horizon:	1,000,000
Replicates:	12
Seed:	42
Metric:	Swap regret
View:	$R / \sqrt{T}$
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
BM-Hedge vs BM-Hedge:	mean = 1.25113, SD = 0.58951, SE = 0.170177
BM-OptHedge vs BM-OptHedge:	mean = 1.19096, SD = 0.271489, SE = 0.078372
Hedge vs Hedge:	mean = 94.2644, SD = 57.7823, SE = 16.6803
Ito-Hedge vs Ito-Hedge:	mean = 1.48221, SD = 0.305174, SE = 0.0880963
OptHedge vs OptHedge:	mean = 5.93767, SD = 2.7385, SE = 0.790536
RM vs RM:	mean = 30.369, SD = 7.24807, SE = 2.09234
SRM vs SRM:	mean = 1.55679, SD = 0.394619, SE = 0.113917

